

General Formulation of the Personnel Scheduling Problem

QI4U in Warsaw 2026

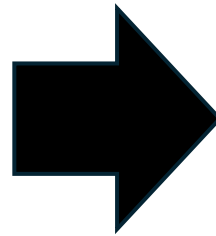
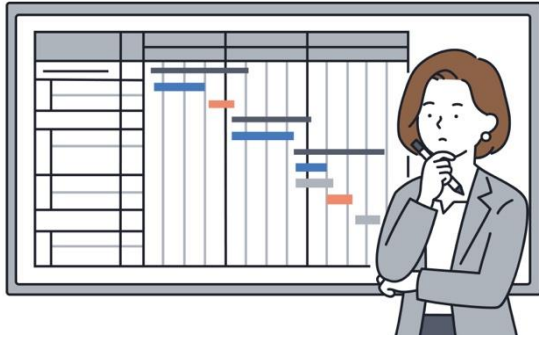
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Ohzeki Group

KAITO YASHIRO

Purpose

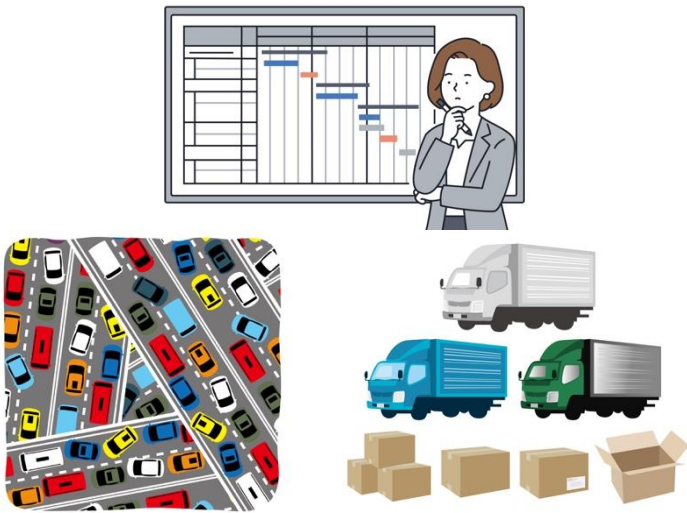
Let's understand quantum annealing and try it out!



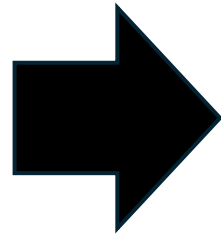
$$\begin{aligned} \Gamma(s) &= \int_0^\infty x^{s-1} e^{-x} dx \quad (s > 0) & \sin x &= 1 - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots & \tan(\alpha \pm \beta) &= \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta} \\ \lim_{x \rightarrow a} \frac{f(x)}{g(x)} &= \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} & \cos \theta &= \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{a_1 b_1 + a_2 b_2}{\sqrt{a_1^2 + a_2^2} \sqrt{b_1^2 + b_2^2}} & \frac{x^2}{a^2} - \frac{y^2}{b^2} &= 1 & R_\theta &= \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \\ (x+y)^n &= \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k & V(x) &= \int_a^b (x-m)^4 f(x) dx & r &= a\theta \\ \int_a^b f(x) dx &= \int_a^b f(a+b-x) dx & \sum_{n=0}^\infty \frac{(-1)^n}{2n+1} &= \frac{\pi}{4} & A &= \begin{pmatrix} a & b \\ c & d \end{pmatrix}, E = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & \zeta(s) &= \sum_{n=1}^\infty \frac{1}{n^s} \\ (\arcsin x)' &= \frac{1}{\sqrt{1-x^2}} & e^{ix} &= \cos x + i \sin x & (a^2+b^2)(x^2+y^2) &\geq (ax+by)^2 & \frac{f(b)-f(a)}{b-a} &= f'(c) \quad (a < c < b) \\ \sqrt{a+b} + 2\sqrt{ab} &= \sqrt{a} + \sqrt{b} & \int f(\sin x, \cos x) &= \int f\left(\frac{2t}{1+t^2}, \frac{1-t^2}{1+t^2}\right) \cdot \frac{2}{1+t^2} dt & \int_{-\infty}^\infty e^{-x^2} dx &= \sqrt{\pi} \\ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} ax+by \\ cx+dy \end{pmatrix} & a^2 &= b^2 + c^2 - 2bc \cdot \cos \alpha & \sin^2 \theta + \cos^2 \theta &= 1 \end{aligned}$$

Purpose

Let's understand quantum annealing and try it out!

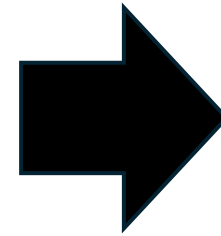


Problem



$$\begin{aligned} \Gamma(\omega) &= \int_0^\infty x^\omega e^{-x} dx \quad (s > 0) & \sin X &= 1 - \frac{1}{3!} X^3 + \frac{1}{5!} X^5 - \frac{1}{7!} X^7 + \dots & \tan(\alpha \pm \beta) &= \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta} \\ \lim_{x \rightarrow a} \frac{f(x)}{g(x)} &= \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} & \cos \theta &= \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{a_1 b_1 + a_2 b_2}{\sqrt{a_1^2 + a_2^2} \sqrt{b_1^2 + b_2^2}} & \frac{x^2}{a^2} - \frac{y^2}{b^2} &= 1 & R_\theta &= \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \\ (x+y)^n &= \sum_{k=0}^n \binom{n}{k} x^k y^{n-k} & V(X) &= \int_a^b (x-m)' f(x) dx & r &= a \cdot \theta \\ \int_a^b f(x) dx &= \int_a^b f(a+b-x) dx & A &= \begin{pmatrix} a & b \\ c & d \end{pmatrix}, E = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & \zeta(s) &= \sum_{n=1}^\infty \frac{1}{n^s} \\ (\arcsin x)' &= \frac{1}{\sqrt{1-x^2}} & \sum_{n=0}^\infty \frac{(-1)^n}{2n+1} &= \frac{\pi}{4} & \Rightarrow A^2 - (a+d)A + (ad-bc)E &= 0 \\ e^{ix} &= \cos x + i \sin x & (a^2+b^2)(x^2+y^2) &\geq (ax+by)^2 & \frac{f(b)-f(a)}{b-a} &= f'(c) \quad (a < c < b) \\ \sqrt{a+b} + 2\sqrt{ab} &= \sqrt{a} + \sqrt{b} & \int f(\sin x, \cos x) dx &= \int f\left(\frac{t}{1+t^2}, \frac{1-t^2}{1+t^2}\right) \frac{2}{1+t^2} dt & \int_{-\infty}^\infty e^{-x^2} dx &= \sqrt{\pi} \\ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} ax+by \\ cx+dy \end{pmatrix} & a^2+b^2+c^2-2bc \cdot \cos \alpha & \sin^2 \theta + \cos^2 \theta &= 1 \end{aligned}$$

Problem formulation



Programming

Agenda

I. Background and Problem Statement

- What is the Personnel Scheduling Problem?
- Problem Background and Method

II. Method and Applications of QUBO Formulation

- Annealing-Based Approach
- Proposed QUBO Formulation

III. Results and Discussion

- Case Study 1: Implementation Results for a Specific Problem

IV. Hands-on Part

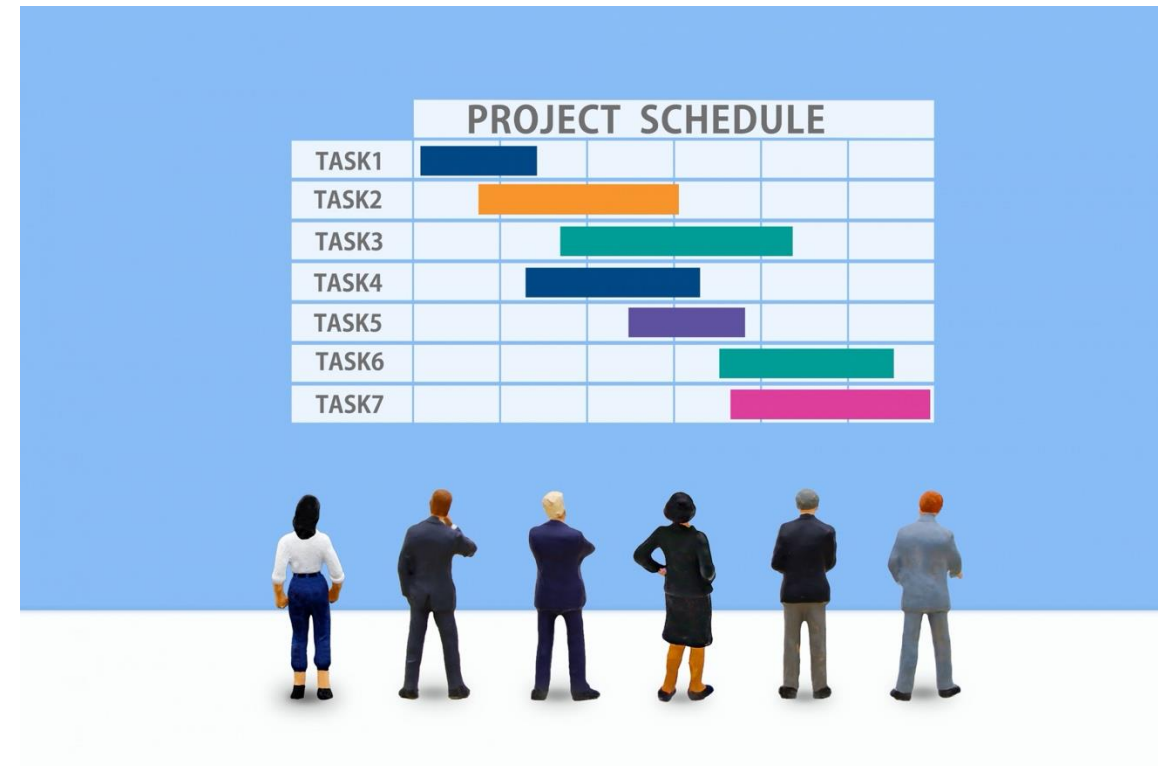
Problem Background

◆ What is the **personnel scheduling problem**?

- The problem of deciding **which worker to assign to which task** when there is a worker, task, or schedule (work period).

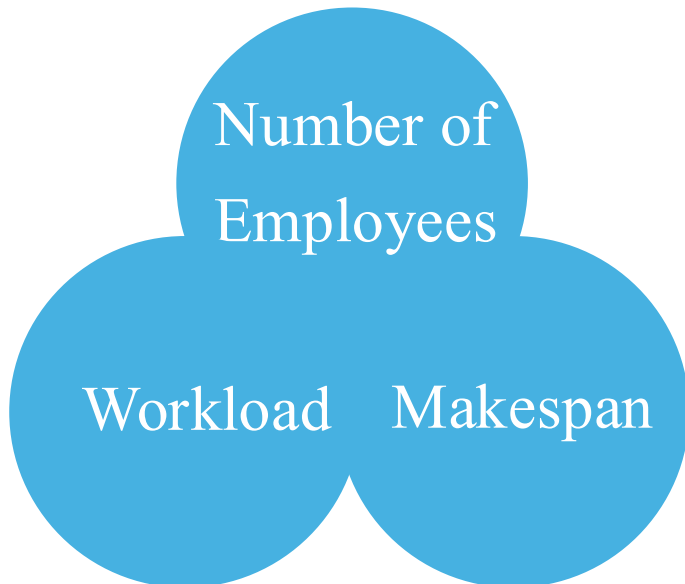
➤ Difficult point

1. As the number of workers and the number of tasks increases, **it becomes difficult to make assignments themselves**.
2. It is difficult to find a exact solution **due to a lot of constraints**.
3. As the size of the problem becomes larger, **the amount of time it takes to allocate is enormous**.



Problem Background

- Personnel scheduling is a highly constrained optimization problem, and **finding exact solutions becomes difficult as the problem size grows.**
- Here, we introduce a **basic formulation** that can be applied generically, without restricting to a specific job type.
- In many formulations, the objective function minimizes the following components:



Problem Background

Which person should be assigned to each task?



Worker



Project Manager

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9
Task 1					 ?		 ?		 ?
Task 2		 ?		 ?	 ?				
Task 3			 ?	 ?				 ?	
Task 4	 ?	 ?			 ?				
Task 5	 ?		 ?	 ?			 ?		
Task 6	 ?					 ?			
Task 7		 ?						 ?	

Problem Background

Which person should be assigned to each task?



Worker



Project Manager

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9
Task 1	 ?	 ?		 ?			 ?		 ?
Task 2				 ?		 ?			
Task 3			 ?						?
Task 4		 ?			 ?				
Task 5			 ?	 ?			 ?		
Task 6			 ?			 ?			
Task 7	 ?				 ?			 ?	 ?

Problem Background

Which person should be assigned to each task?



Worker



Project Manager

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9
Task 1	 ?			 ?			 ?		 ?
Task 2	 ?			 ?					 ?
Task 3	 ?		 ?				 ?		
Task 4									 ?
Task 5									
Task 6								 ?	
Task 7	 ?	 ?		 ?	 ?			 ?	 ?

Problem Background

Which person should be assigned to each task?



Worker



Project Manager

	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9
Task 1			 ?	 ?			 ?		 ?
Task 2				 ?					 ?
Task 3	 ?		 ?				 ?		
Task 4					 ?			 ?	 ?
Task 5			 ?	 ?		 ?			
Task 6								 ?	
Task 7		 ?			 ?		 ?	 ?	 ?

Problem Background

Which person should be assigned to each task?



Worker



Project Manager

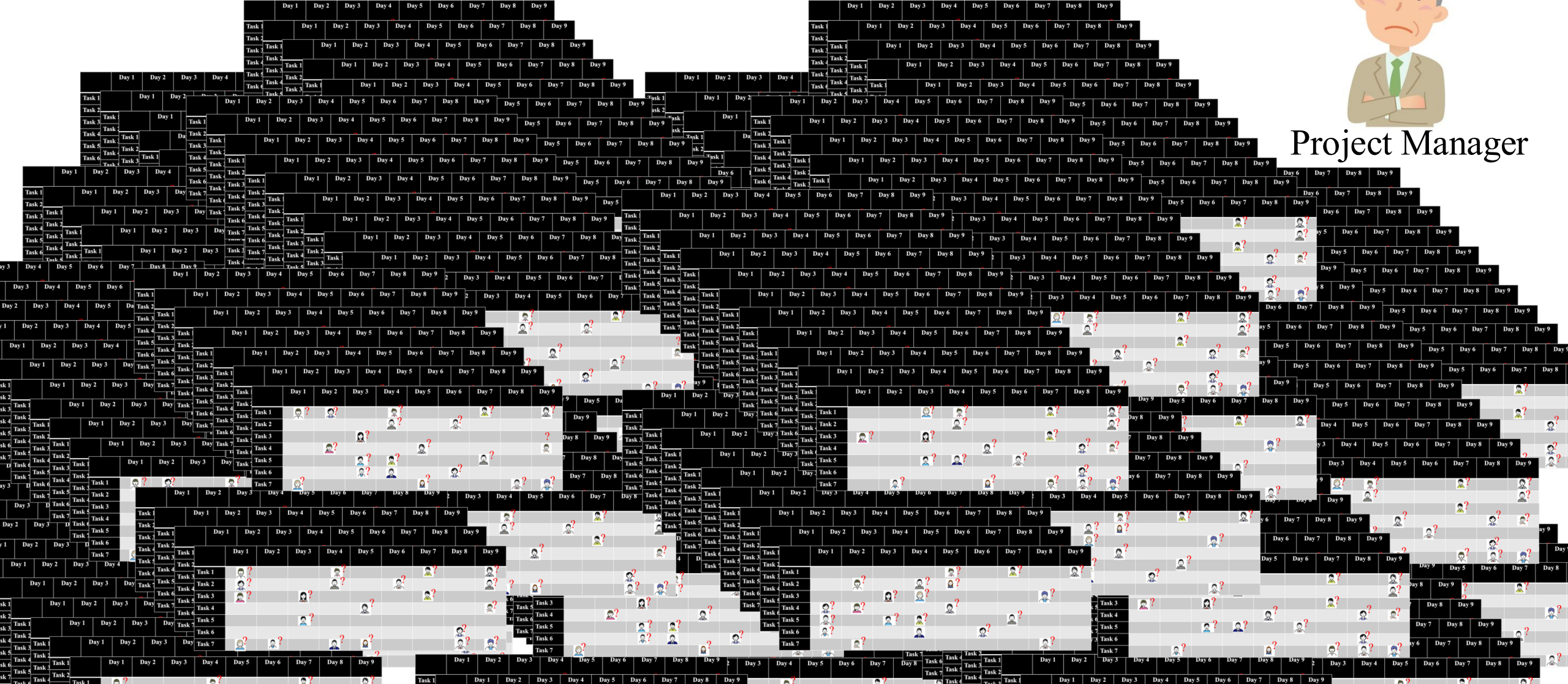
	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9
Task 1		 ?		 ?			 ?		
Task 2				 ?		 ?		 ?	 ?
Task 3	 ?		 ?				 ?		
Task 4								 ?	 ?
Task 5	 ?	 ?	 ?	 ?		 ?	 ?		
Task 6							 ?		
Task 7	 ?			 ?	 ?				

Problem Background

There are countless possible combinations...



Project Manager

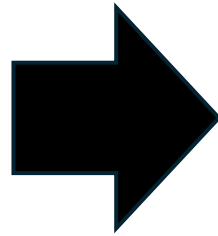
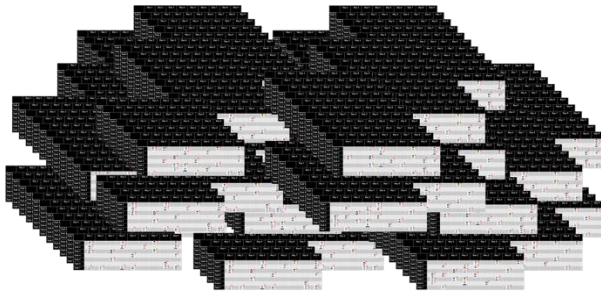


Problem Background



Project Manager

Quantum
Annealing



	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9
Task 1									
Task 2									
Task 3									
Task 4									
Task 5									
Task 6									
Task 7									

Technically, sampling can be performed in a short time.

It can obtain the multiple solutions!

Problem Background

◆ Problem Setup

- Assign tasks that are easy for workers to perform or that they already have experience with.
- Given the number of workers, the number of tasks, and the assignable time periods in advance, determine which worker is assigned to which task and at what time.
- A formulation that respects workers' preferences.

Is it possible to reduce this problem to a QUBO formulation and find the multiple solutions?

Method

◆ Annealing-Based Approach

→ QUBO (**q**uadratic **u**nconstrained **b**inary **o**ptimization)

A method for obtaining an approximate solution to a model problem.

$$\begin{array}{c} \text{QUBO} \\ \hline \min_{x \in \{0,1\}^N} \sum_{i \leq j}^N Q_{ij} x_i x_j \end{array}$$

Method

What do you define as the variable x ?

Selecting :1

Not Selecting :0

Delivering :1

Not Delivering :0

Assigning :1

Not Assigning :0

Right :1

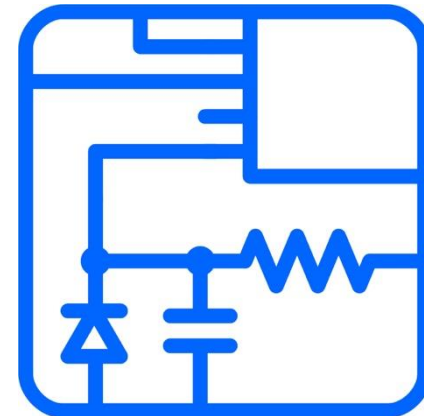
Left :0

ON :1

OFF :0

Going :1

Not Going :0



Applications of QUBO Formulation

◆ Preparation of the Proposed QUBO Formulation

➤ Variable definition

- $x_{i,k,t} = \begin{cases} 1 & \text{If Worker } i \text{ is assigned to task } k \text{ at time } t, \\ 0 & \text{Others.} \end{cases}$

	Day 1	Day 2	Day 3	Day 4	Day 5
Task 1	0	 1	0	 1	0
Task 2	0	0	0	 1	0
Task 3	 1	0	 1	0	0


 $x_{i,3,3} = 1$

Applications of QUBO Formulation

◆ Proposed QUBO Formulation

QUBO

$$\min_{x \in \{0,1\}^N} \sum_{i \leq j}^N Q_{ij} x_i x_j$$

$$E = -\alpha \sum_{i=1}^I \sum_{k=1}^K \sum_{t=1}^T C_{i,k} x_{i,k,t} - \beta \sum_{i=1}^I \sum_{k=1}^K \sum_{t=1}^T T_{i,k} x_{i,k,t}$$

Objective Term

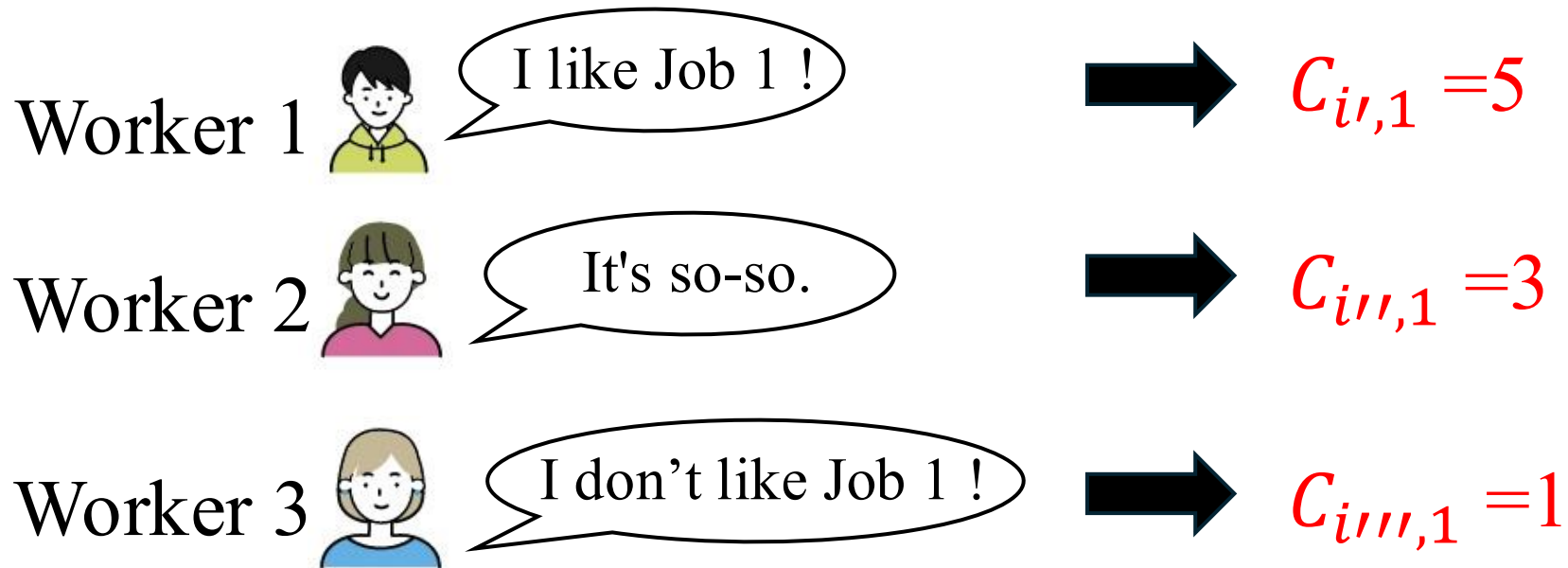
- i. Maximizing Interest ii. Maximizing Experience

Applications of QUBO Formulation

◆ Preparation of the Proposed Proposed QUBO Formulation

➤ Definition of constants

- $C_{i,k} \in \mathbb{N}$: **Level of interest** in task k for worker i . $\rightarrow (1,2,3,4,5)$

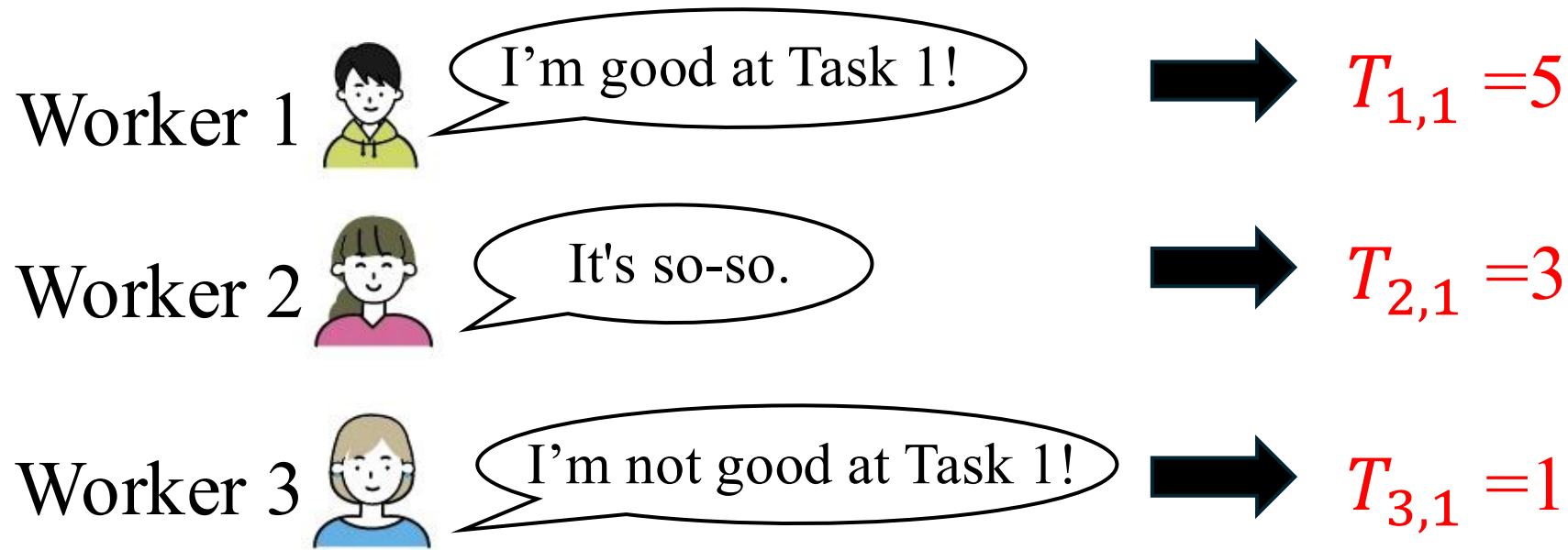


Applications of QUBO Formulation

◆ Preparation of the Proposed Proposed QUBO Formulation

➤ Definition of constants

- $T_{i,k} \in \mathbb{N}$: **Level of experience** in task k for worker i . $\rightarrow (1,2,3,4,5)$

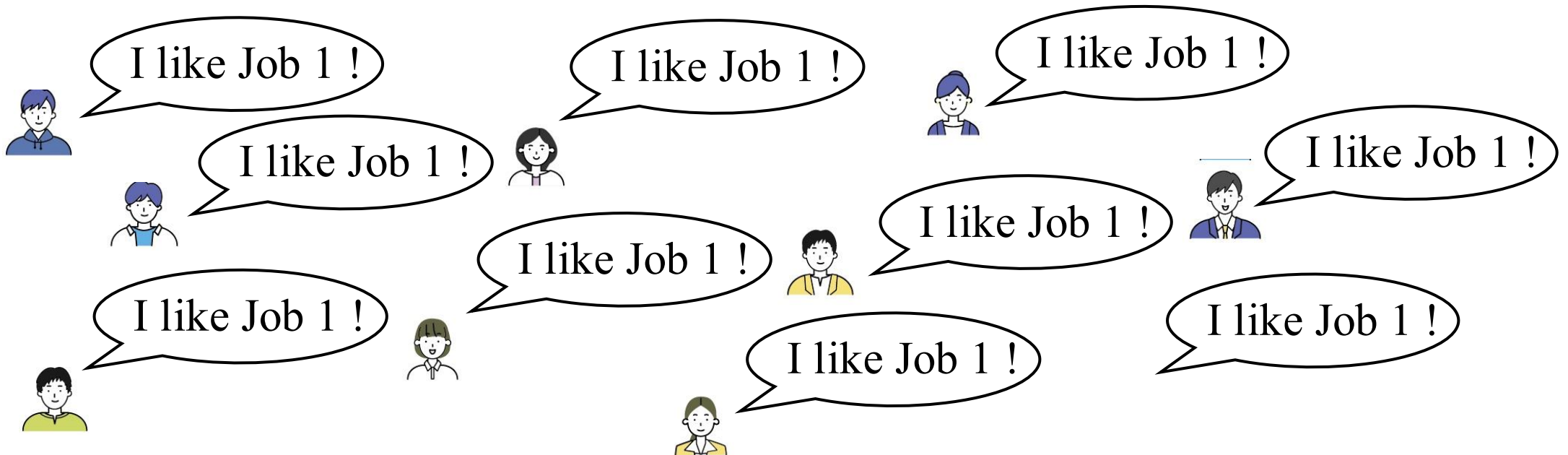


Applications of QUBO Formulation

◆ Preparation of the Proposed Proposed QUBO Formulation

$$E = -\alpha \sum_{i=1}^I \sum_{k=1}^K \sum_{t=1}^T C_{i,k} x_{i,k,t} - \beta \sum_{i=1}^I \sum_{k=1}^K \sum_{t=1}^T T_{i,k} x_{i,k,t}$$

i. Maximizing Interest and Experience



Applications of QUBO Formulation

◆ Preparation of the Proposed Proposed QUBO Formulation

$$E = -\alpha \sum_{i=1}^I \sum_{k=1}^K \sum_{t=1}^T C_{i,k} x_{i,k,t} - \beta \sum_{i=1}^I \sum_{k=1}^K \sum_{t=1}^T T_{i,k} x_{i,k,t}$$

i. Maximizing Interest and Experience

	Day 1	Day 2	Day 3	Day 4	Day 5
Task 1	     	  			
Task 2	     	  			
Task 3					

Bad Situation



Constraints are required.



Project Manager

Applications of QUBO Formulation

◆ Proposed QUBO Formulation

Constraint Term

$$+ \lambda_1 \sum_{i=1}^I \sum_{t=1}^T \left(\sum_{k=1}^K x_{i,k,t} - 1 \right)^2$$

iii. No Multitasking
(Hard Constraint)

➤ Variable definition

- $x_{i,k,t} = \begin{cases} 1 & : \text{Worker } i \text{ is assigned to task } k \text{ at time } t, \\ 0 & : \text{Others.} \end{cases}$

✓ Using a penalty term, performing multiple tasks simultaneously is penalized.

Applications of QUBO Formulation

$$+ \lambda_1 \sum_{i=1}^I \sum_{t=1}^T \left(\sum_{k=1}^K x_{i,k,t} - 1 \right)^2$$

Worker 1		$\sum_1^K x_{1,k,1}$	$\sum_1^K x_{1,k,2}$	$\cdots$	$\sum_1^K x_{1,k,6}$
Worker 2		$\sum_1^K x_{2,k,1}$	$\sum_1^K x_{2,k,2}$	$\cdots$	$\sum_1^K x_{2,k,6}$
Worker 3		$\sum_1^K x_{3,k,1}$	$\sum_1^K x_{3,k,2}$	$\cdots$	$\sum_1^K x_{3,k,6}$
		Time 1	Time 2		Time 6

Sum of tasks worker3 do at time6.

Applications of QUBO Formulation

$$+ \lambda_1 \sum_{i=1}^I \sum_{t=1}^T \left(\sum_{k=1}^K x_{i,k,t} - 1 \right)^2$$

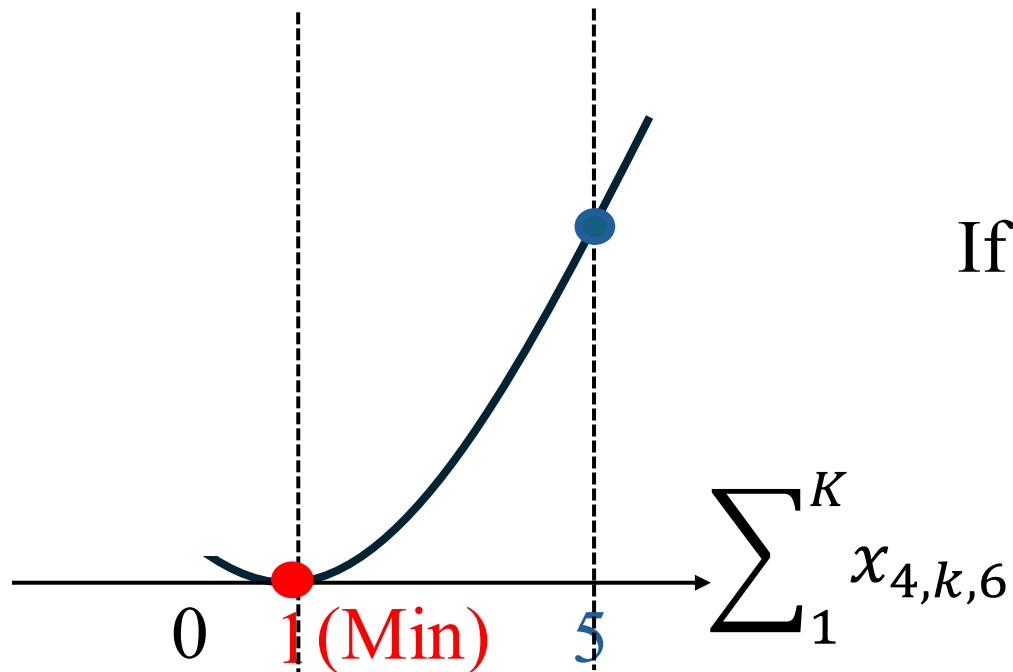
Worker 1		$\sum_1^K x_{1,k,1}$	$\sum_1^K x_{1,k,2}$	$\cdots$	$\sum_1^K x_{1,k,6}$
Worker 2		$\sum_1^K x_{2,k,1}$	$\sum_1^K x_{2,k,2}$	$\cdots$	$\sum_1^K x_{2,k,6}$
Worker 3		$\sum_1^K x_{3,k,1}$	$\sum_1^K x_{3,k,2}$	$\cdots$	$\sum_1^K \cancel{x_{3,k,6}}$
		Time 1	Time 2		Time 6

Worker 3 has 7 tasks at time 6...

Applications of QUBO Formulation

$$+ \lambda_1 \sum_{i=1}^I \sum_{t=1}^T \left(\sum_{k=1}^K x_{i,k,t} - 1 \right)^2$$

Quadratic Function



The sum of tasks assigned to Worker 4 at Time 6 equals 1.

If $\sum_{k=1}^K x_{4,k,6} = 1$, $\left(\sum_{k=1}^K x_{4,k,6} - 1 \right)^2 = 0$

If $\sum_{k=1}^K x_{4,k,6} = 5$, $\left(\sum_{k=1}^K x_{4,k,6} - 1 \right)^2 = 16$

Applications of QUBO Formulation

Which is the best for Quantum annealing?

If $\sum_1^K x_{4,k,6} = 1, \left(\sum_1^K x_{4,k,6} - 1\right)^2 = 0$



The sum of tasks assigned to
Worker 4 at Time 6 equals 1.

If $\sum_1^K x_{4,k,6} = 5, \left(\sum_1^K x_{4,k,6} - 1\right)^2 = 16$



The sum of tasks assigned to
Worker 4 at Time 6 equals 5.



Applications of QUBO Formulation

Which is the best for Quantum annealing?

$$\text{If } \sum_1^K x_{4,k,6} = 1, \quad \left(\sum_1^K x_{4,k,6} - 1 \right)^2 = 0$$



The sum of tasks assigned to
Worker 4 at Time 6 equals 1.

$$\text{If } \sum_1^K x_{4,k,6} = 5, \quad \left(\sum_1^K x_{4,k,6} - 1 \right)^2 = 16$$



The sum of tasks assigned to
Worker 4 at Time 6 equals 5.



Applications of QUBO Formulation

◆ Preparation of the Proposed Proposed QUBO Formulation

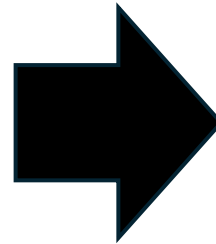
$$E = -\alpha \sum_{i=1}^I \sum_{k=1}^K \sum_{t=1}^T C_{i,k} x_{i,k,t} - \beta \sum_{i=1}^I \sum_{k=1}^K \sum_{t=1}^T T_{i,k} x_{i,k,t} + \lambda_1 \sum_{i=1}^I \sum_{t=1}^T \left(\sum_{k=1}^K x_{i,k,t} - 1 \right)^2$$

Constraint Term

i. Maximizing Interest and Experience

iii. No Multitasking
(Hard Constraint)

	Day 1	Day 2	Day 3
Task 1			
Task 2			
Task 3			



	Day 1	Day 2	Day 3
Task 1			
Task 2			
Task 3			

Applications of QUBO Formulation

Other constraints

- Required Workload Satisfaction
- Work continuously
- Consideration of Task
- Consider work hours of workers

Applications of QUBO Formulation

◆ Proposed QUBO Formulation

$$\begin{aligned}
 E = & \underbrace{-\alpha \sum_{i=1}^I \sum_{k=1}^K \sum_{t=1}^T C_{i,k} x_{i,k,t}}_{\text{Objective Term i. Maximizing Interest}} - \underbrace{\beta \sum_{i=1}^I \sum_{k=1}^K \sum_{t=1}^T T_{i,k} x_{i,k,t}}_{\text{Objective Term ii. Maximizing Experience}} \\
 & + \underbrace{\lambda_1 \sum_{i=1}^I \sum_{t=1}^T \left(\sum_{k=1}^K x_{i,k,t} - 1 \right)^2}_{\text{Constraint Term iii. No Multitasking (Hard Constraint)}} + \underbrace{\lambda_2 \sum_{k=1}^K \left(\sum_{i=1}^I \sum_{t=1}^T x_{i,k,t} - W_k \right)^2}_{\text{Constraint Term iv. Required Workload Satisfaction (Hard Constraint)}} + \underbrace{\lambda_3 \sum_{i=1}^I \sum_{k=1}^K \sum_{t=1}^{T-1} (x_{i,k,t} - x_{i,k,t+1})^2}_{\text{Constraint Term v. Ensuring Continuity (Soft Constraint)}} \\
 & + \underbrace{\lambda_4 \sum_{k=1}^K \sum_{1 \leq i < j \leq I} \sum_{t=1}^T (x_{i,k,t} - x_{j,k,t})^2}_{\text{Constraint Term vi. Consistent Scheduling of the Same Task (Soft Constraint)}} + \underbrace{\lambda_5 \sum_{k=1}^K \sum_{i=1}^I \sum_{t=1}^T (t \cdot \text{Pr}_k) x_{i,k,t}}_{\text{Constraint Term vii. Consideration of Task Priority (Soft Constraint)}} + \underbrace{\lambda_6 \sum_{i=1}^I \sum_{t=1}^T \sum_{k=1}^K (1 - D_{i,t}) x_{i,k,t}}_{\text{Constraint Term viii. Consider work hours (Hard Constraint)}}
 \end{aligned}$$

Results and Discussion (1/7)

◆ Case Study 1: Implementation Results for a Specific Problem

➤ Initial Condition 1

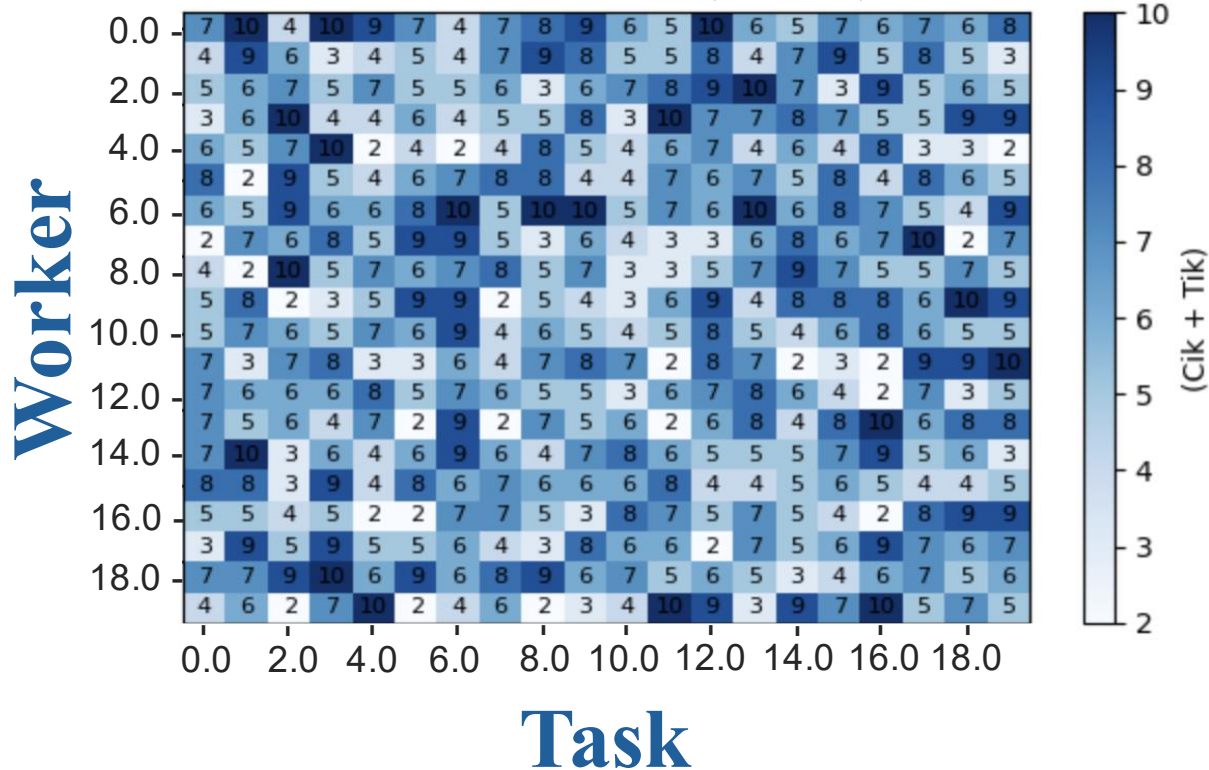
Parameter	Symbols / Values
Total Number of Workers, Tasks, and Time Periods	$I = 20, K = 20, T = 12$
Priority Score	$Pr_k \in \{1, 2, 3, 4, 5\}$
Interest and Experience Levels	$C_{i,k}, T_{i,k} \in \{1, 2, 3, 4, 5\}$
Weight Parameters α, β	$\alpha = \beta = 1$
Constraint Weight Parameter λ	$[\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5, \lambda_6] = [4, 11, 4, 0.6, 0.2, 20]$
Solver Used	openjij, SASampler (num_reads=1000)

Results and Discussion (3/7)

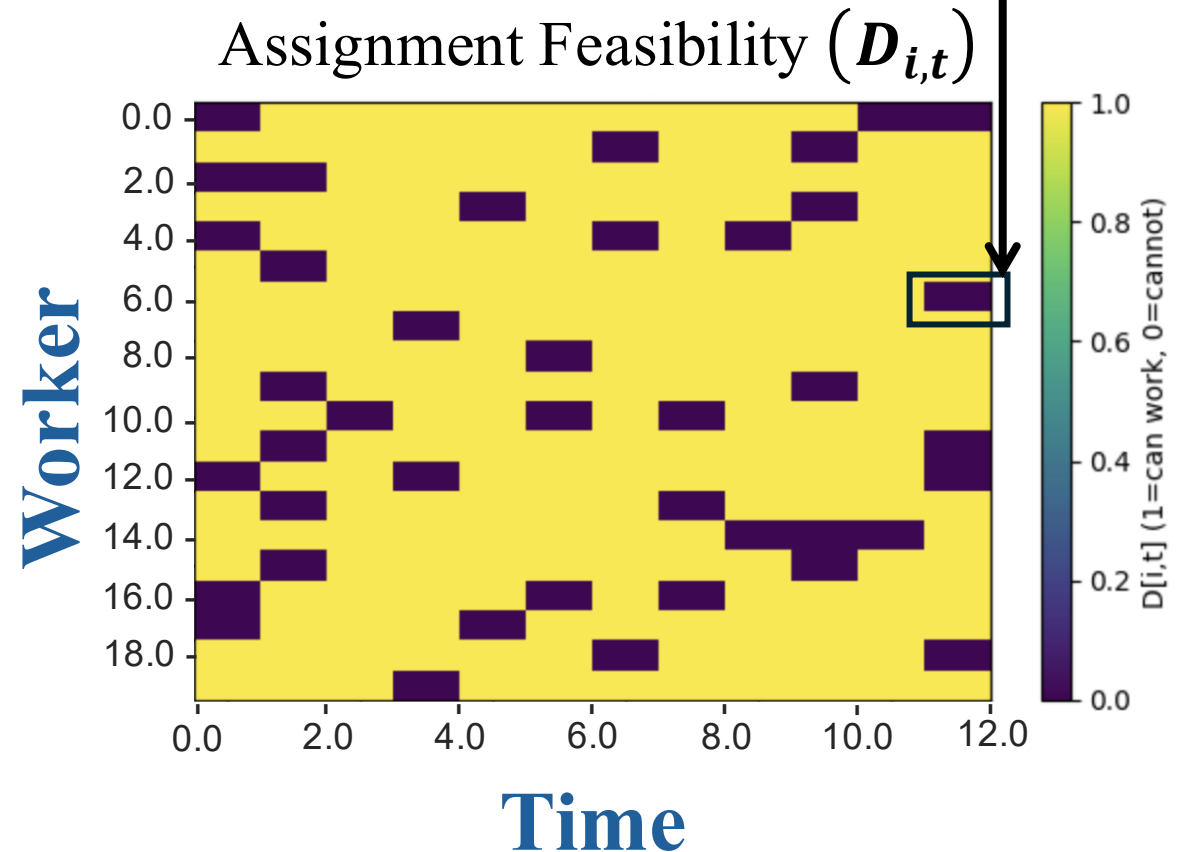
◆ Case Study 1: Implementation Results for a Specific Problem

➤ Initial Condition 2

Distribution of $(C_{i,k} + T_{i,k})$



Unavailable Time Slots for Assignment



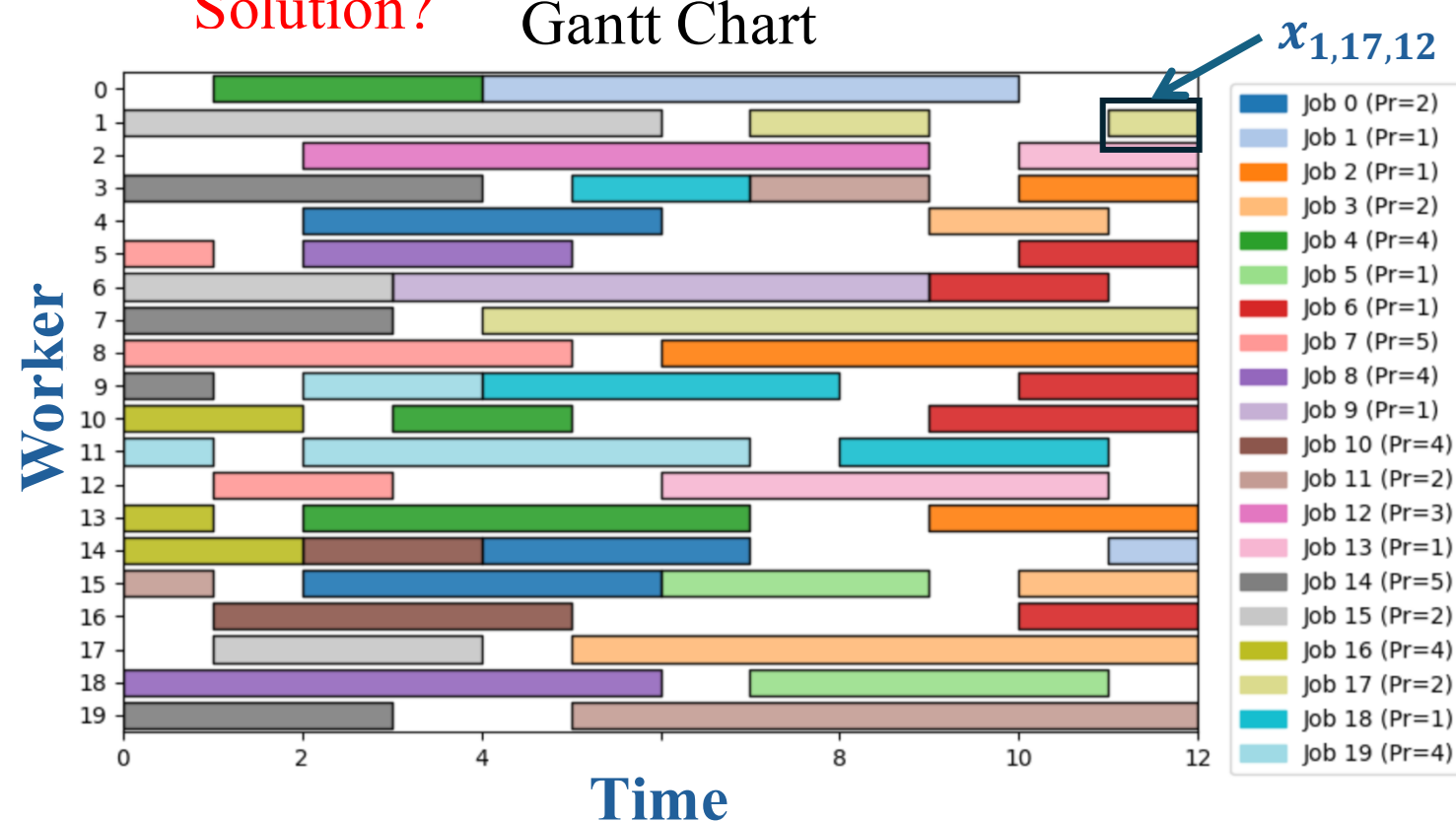
Results and Discussion (4/7)

◆ Case Study 1: Implementation Results for a Specific Problem

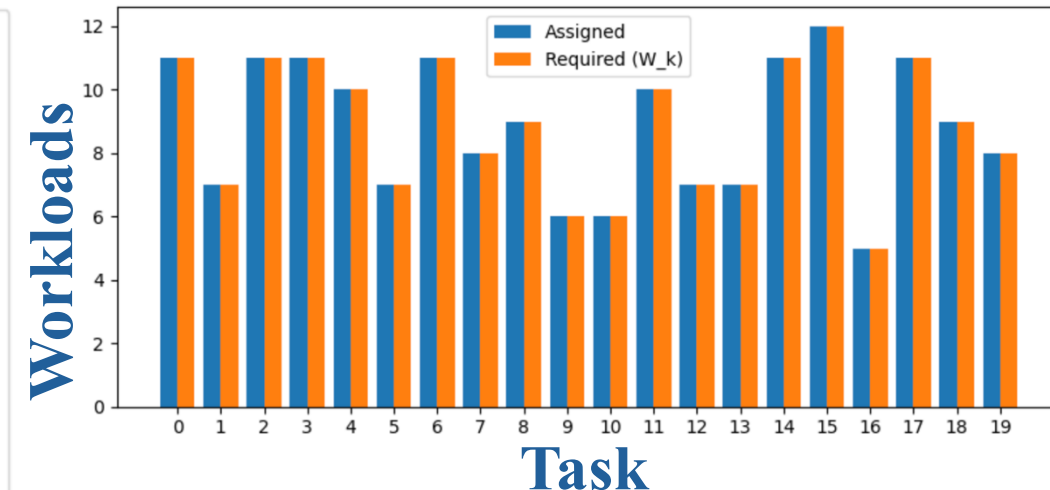
➤ Implementation Results

Is It Possible to Formulate the Problem as a QUBO Model and Obtain the Optimal Solution?

Gantt Chart



$x_{1,17,12} = 1$ Feasibility of Workload Satisfaction



Constraint (iii): Prohibition of Multitasking

Constraint (iv): Satisfaction of Required Workload

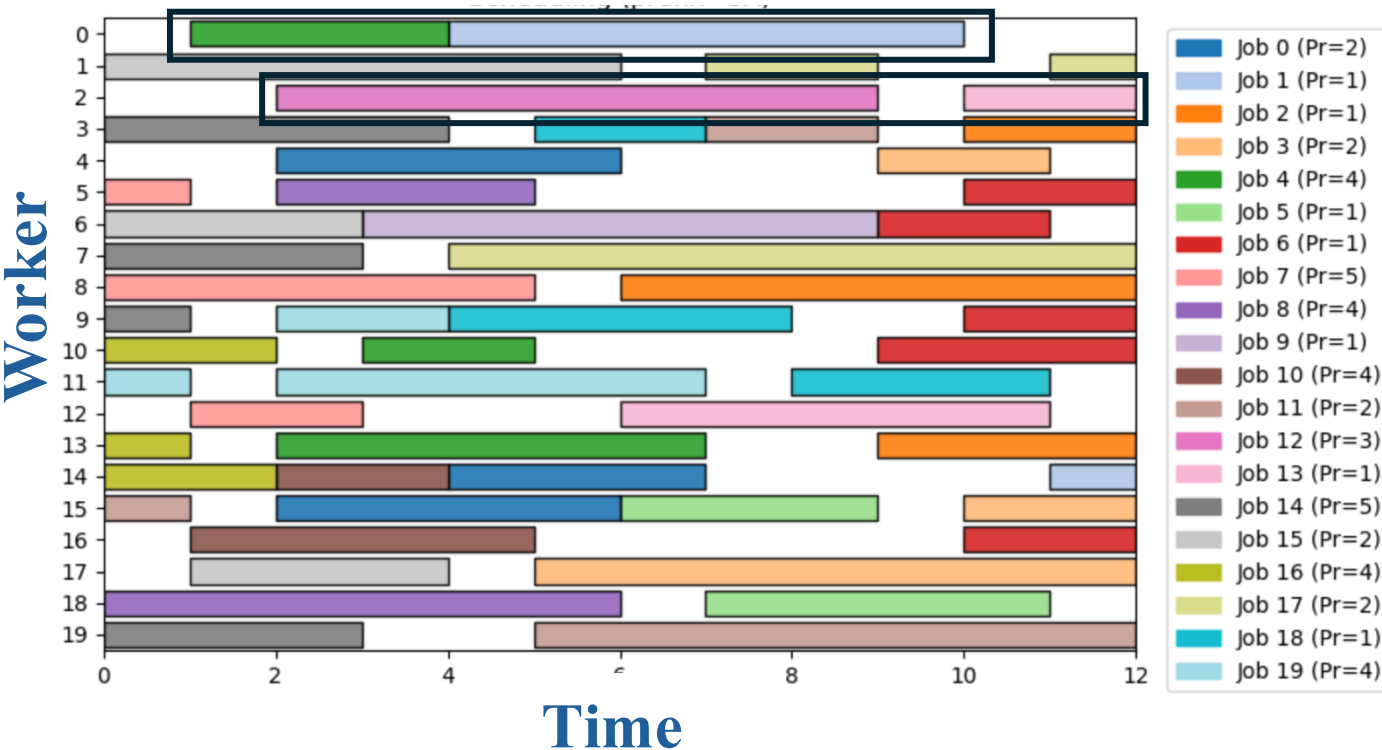
Results and Discussion (5/7)

◆ Case Study 1: Implementation Results for a Specific Problem

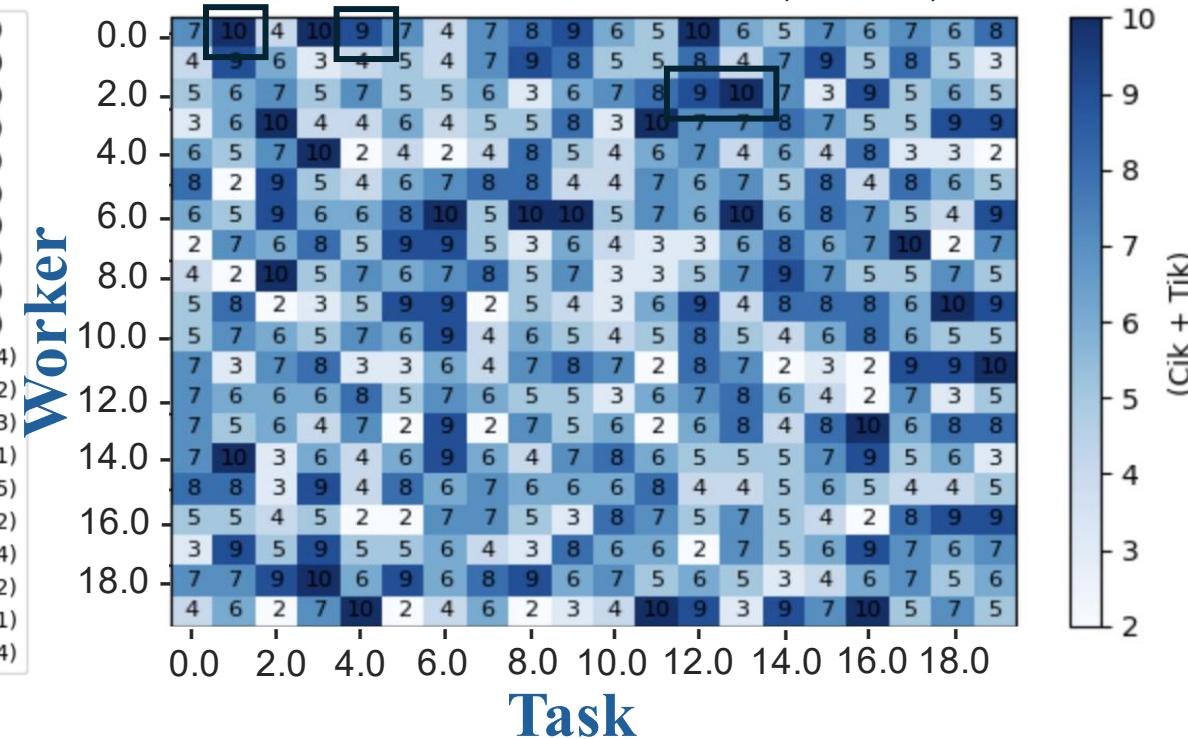
➤ Implementation Results

➡ Maximizing the Interest Score / Experience Score

Gantt Chart



Distribution of $(C_{i,k} + T_{i,k})$

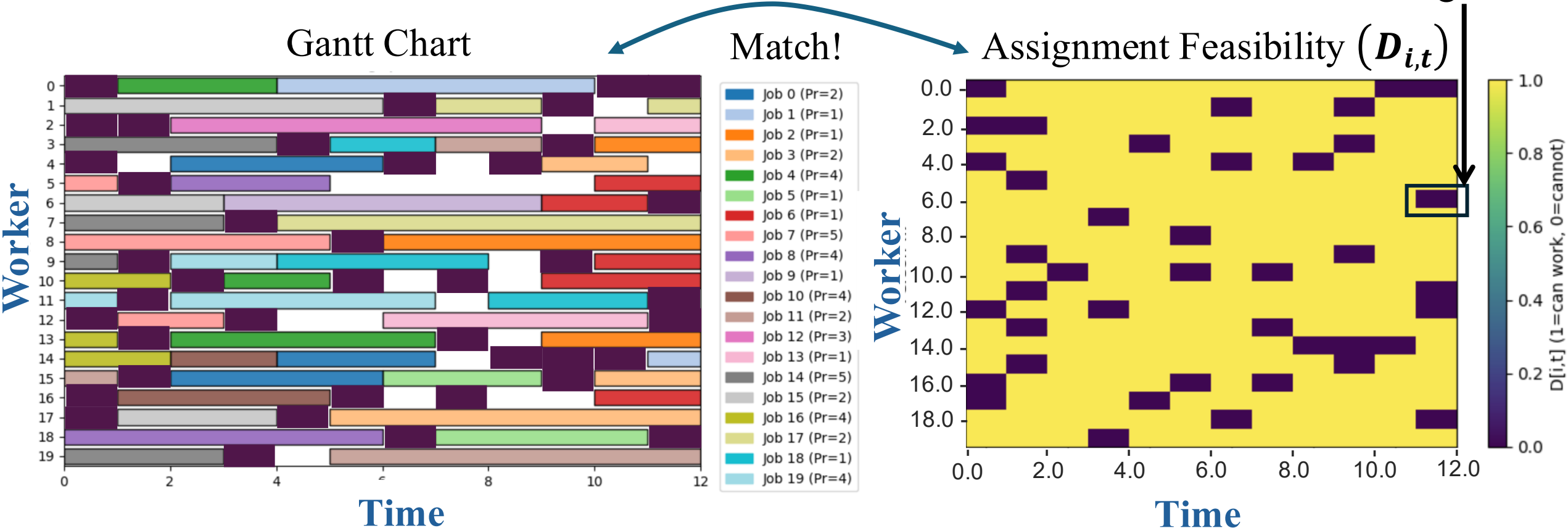


Results and Discussion (6/7)

◆ Case Study 1: Implementation Results for a Specific Problem

➤ Implementation Results

➡ Constraint (viii): Availability of Working Time Unavailable Time Slots for Assignment



Results and Discussion (7/7)

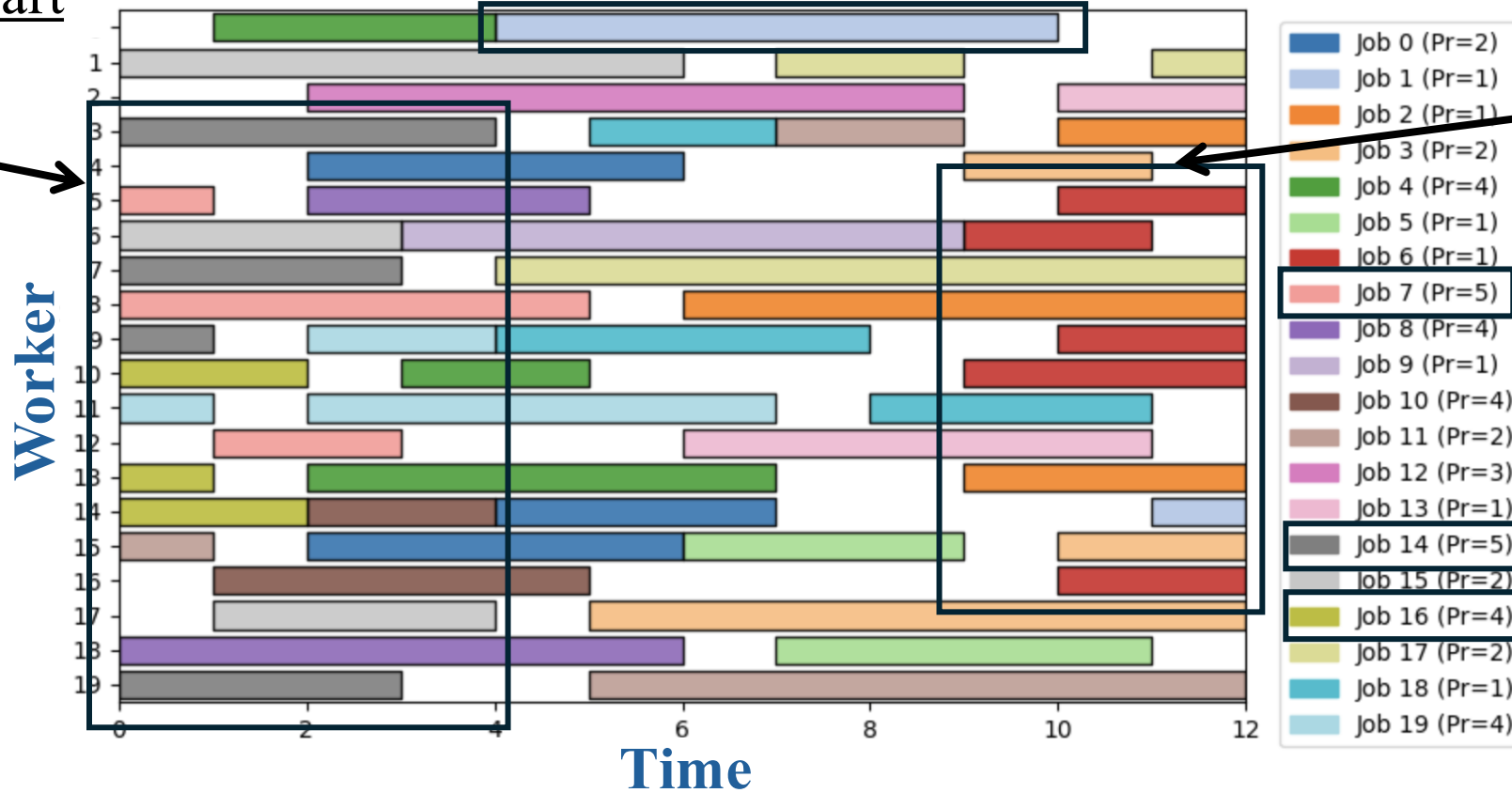
◆ Case Study 1: Implementation Results for a Specific Problem

➤ Implementation Results

➡ Satisfaction of Constraints (v), (vi), and (vii) ✓ Constraint (v): Ensuring Continuity

Gantt Chart

✓ Constraint (vii): High-Priority Tasks Are Assigned Earlier



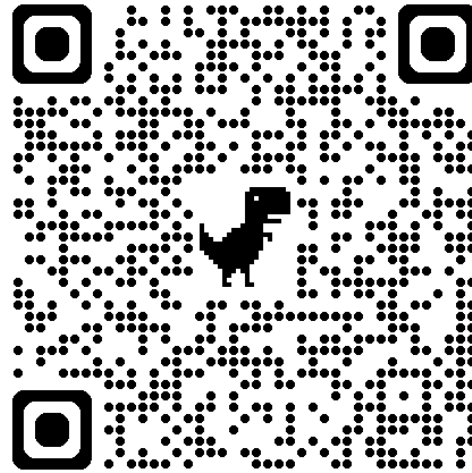
Let's try solving the problem!

Please open the Google colab or your own coding environment.

<https://bit.ly/4rKFDaB>

More information

Scheduling Problem:
A More Realistic Formulation !



<https://bit.ly/4cXzmnu>

Dziękuję !